

Research on the Application of Data Warehouse in Education Evaluation System

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ABSTRACT

As an efficient data management and analysis technology, the data warehouse realizes centralized storage and management of data by integrating data scattered in educational institutions. It can dig out the laws and trends hidden behind the data and provide more accurate and comprehensive data support for educational evaluation. In the educational evaluation system, the application of data warehouses is embodied in data integration and management, multidimensional evaluation, personalized learning support, and educational decision optimization. Through a multidimensional evaluation model, the school can comprehensively evaluate students' academic performance and comprehensive quality and provide strong support for personalized teaching. Data warehouses can also record and analyze students' learning trajectory and provide customized learning materials and exercises for each student, which significantly improves learning efficiency.

KEYWORDS

Data Warehouse, Educational Evaluation System, Applied Analysis, Empirical Research

INTRODUCTION

In today's rapidly developing education field, the educational evaluation system, as an important tool to measure the quality of education, guide education reform, and improve educational effects, is facing unprecedented challenges and opportunities (Bozkurt et al., 2020). As efficient data management and analysis technologies, data warehouses in education evaluation systems are gradually becoming a hot topic in the field of education (Zhao et al., 2021).

The core of the education evaluation system is to provide educators, learners and managers with scientific and objective evaluation results and decision-making bases by collecting, sorting out, and analyzing multidimensional data, such as students' academic performance, comprehensive quality, and behavior performance (Xin et al., 2022). However, the traditional education evaluation system often has some problems, such as scattered data, low processing efficiency, and isolated information islands, which lead to inaccurate and not comprehensive evaluation results and struggle to provide strong support for educational decision-making (Rudolph et al., 2023). Therefore, how to effectively integrate and utilize massive educational data has become the key to improving the efficiency of the educational evaluation system.

As a subject-oriented, integrated, relatively stable, and time-varying form of data collection, a data warehouse is often used to support management decision-making. In the educational evaluation system, a data warehouse can not only store and manage the data scattered in all corners of educational institutions but also realize the integration and unification of data. It can also uncover the laws and

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trends hidden behind the data through powerful data processing and analysis capabilities and provide more accurate and comprehensive data support for educational evaluation (Tattersall, 2021). In addition, its application in the educational evaluation system has important practical significance. On one hand, it can help educators to master students' learning situations and growth trajectories more accurately and provide strong support for personalized teaching. On the other hand, it can also provide more comprehensive and in-depth educational data for educational administrators and help them better understand the current educational situation and formulate more scientific and reasonable educational policies and plans (Guorui, 2023).

Everything has two sides, and the application of data warehouse in education evaluation systems is also facing many challenges: for example, how to ensure the security and privacy of data, how to effectively process and analyze massive data, and how to build an efficient data warehouse system. These problems require conducting in-depth research and exploration (Ahmadi, 2023; Kahn et al., 2022). Therefore, this paper aims to explore the application of data warehouses in educational evaluation systems and analyze their advantages in data processing, analysis, and decision support, as well as their challenges and solutions. It is hoped that the research in this paper can provide some theoretical reference and practical guidance for the application of data warehouses in education, promote the continuous optimization and upgrading of the educational evaluation system, and contribute to improving education quality and promoting education equity.

LITERATURE REVIEW

Garani et al. (2019) proposed a data warehouse schema modeling method that integrated temporal and spatial data into a general data warehouse framework in a unified way. With the growth of data and sharing, the integration of temporal and spatial data becomes more important. The purpose of this research is to promote the understanding, query and management of spatiotemporal data for online analytical processing. The proposed new spatiotemporal data warehouse architecture extended online analytical processing queries to support spatial and temporal queries. A case study was developed and implemented for the telecommunications industry.

Wahono and Ali (2021) studied the role of data warehouses, software, and brain ware in decision-making. In decision-making activities, managers can simplify and speed up the response time to environmental changes with the help of information systems. The function of software (DSS application) is to provide commands to the computer to make it run in the best way—using the user or the user's wishes to run synchronously. At the same time, the data warehouse is the centralized and integrated storage of company or organization data, and its arrangement greatly supports the decision-making process by providing input for application software. Application software needs the data warehouse to run its processes accurately.

Nambiar and Mundra (2022) examined two popular data management systems in the field of big data analysis—namely, the data warehouse and data lake, which serve as platforms to accumulate big data generated and used by organizations. The two systems are different in characteristics and applications. This paper summarizes in detail the functions of data warehouses and data lakes in modern enterprises' data management. The definition, characteristics and related work of each data management framework are introduced in detail.

Rudniy (2022) described the design and construction process of a data warehouse for an online learning platform, which uses three main technologies: Microsoft SQL Server, MongoDB and Apache Hive. The case study also showed the value of evidence-centered design principles for data warehouse design. In this work, data was processed, transformed and stored in three versions of a data warehouse to find a better and more suitable platform. The results showed that Apache Hive had the best processing time, followed by SQL Server and MongoDB. In analyzing queries, SQL Server performed best, followed by MongoDB and Hive. Suggestions for future research included

trying data warehouse design in an environment involving larger datasets and finer operations and ensuring that attention is paid to the sustainability of operations.

With the development of technology, data warehouses have been applied to medical treatment, architecture, commerce and other fields. Among them, some scholars in the field of education have also conducted related research.

Thitirath (2019) studied the performance evaluation of schools in northern Thailand in the data warehouse and business intelligence system (DWBIS) for education quality assurance. This project was used to determine evaluation indicators and standards to evaluate system efficiency to meet the management objectives of educational institutions. The results of this research can be applied to schools interested in improving the DWBIS quality evaluation system. Through the improvement and guidance of empirical data, the DWBIS system can provide efficiency and effectiveness that meet the requirements of educational institutions.

Data warehouse tools and technologies cannot handle the data load and analysis process to provide meaningful information for senior managers. Santoso (2017) implemented big data technology to expand existing data warehouse solutions. This paper discussed how to use a data warehouse to implement big data technology to support the decision-making process. In this framework, Hadoop was used as a big data analysis tool for data intake/temporary storage.

Edastama et al. (2021) focused on how to collect operational data in a data warehouse, then apply data mining technology to analyze them. This research produced a comprehensive data warehouse with a web-based information reporting application. In addition, data mining technology was used to analyze the created data warehouse. The result of applying data mining was describing the characteristic pattern of students who take part in some professional courses.

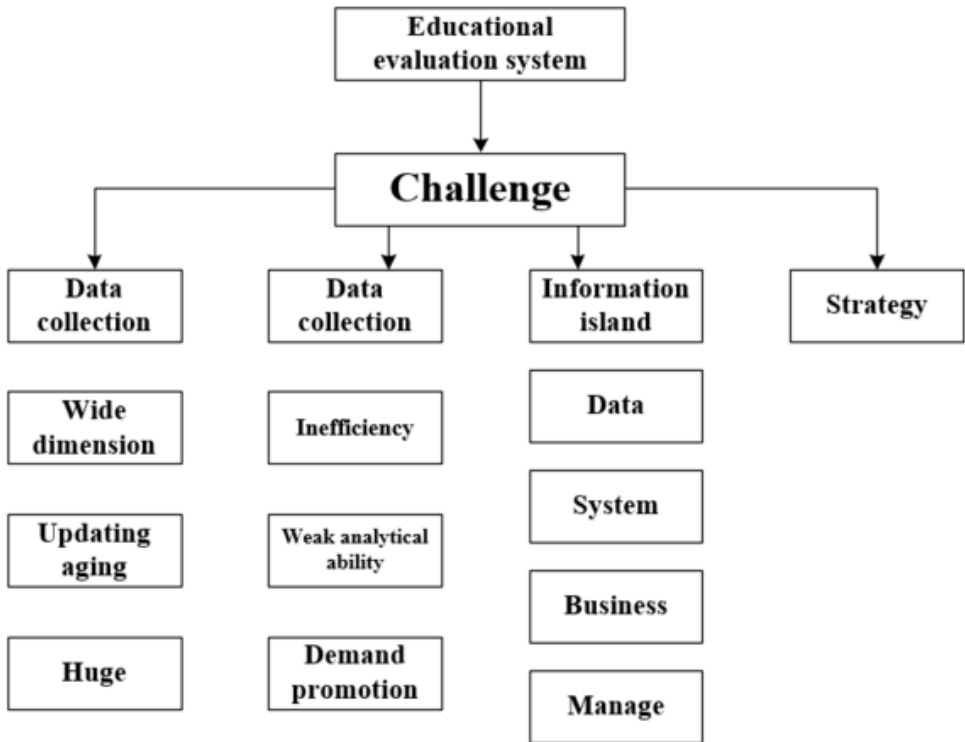
Gladić and Petrovački's (2021) experience of using data warehouse solutions in complex business organizations can also be applied to educational organizations. The challenge of the epidemic situation forced teachers to change their teaching mode and make a comprehensive transition to online learning. In order to evaluate the teaching process and compare its results in traditional and remote environments, data from different sources were analyzed. The authors introduced a pilot project of data warehouse solutions, which was used to monitor students' fulfillment of course requirements in traditional and distance learning environments.

MATERIALS AND METHODS

Present Situation and Challenges of the Educational Evaluation System

With the rapid development of information technology and the explosive growth of educational data, the traditional educational evaluation system is facing various challenges shown in Figure 1.

Figure 1. Challenges faced by the education evaluation system



First of all, from the perspective of data collection, the education evaluation system needs to collect and process various types of data from multiple channels. These data mainly include students' grades, comprehensive quality evaluation, behavior records, teachers' evaluations, parents' feedback, and so on. Due to the problems of inconsistent data formats, uneven data quality and untimely data updates, the traditional education evaluation system often has difficulties with data collection. This not only affects the accuracy and integrity of the data but also limits the objectivity and scientificity of the evaluation results (Wu et al., 2024).

Secondly, in terms of data processing and analysis, the traditional educational evaluation system often adopts simple statistical methods or manual judgment. This method is not only inefficient but ineffective at discovering the deep-seated laws and trends hidden behind the data. With the continuous development of big data technology, our system needs more efficient and intelligent data processing and analysis methods to support more complex and detailed evaluation requirements.

In addition, the traditional education evaluation system also has serious problems with information islands. Information islands refer to computer application systems that are functionally unrelated and not mutually supportive; Information is not shared or exchanged, and it is disconnected from business processes and applications. Due to the imperfect data sharing mechanism between different educational institutions and different departments, a large number of valuable data are isolated and cannot be used effectively. This not only wastes resources but also limits the overall efficiency of the education evaluation system (Liu, 2021).

In addition to the above challenges, the traditional educational evaluation system also faces the problem of insufficient decision support. Due to the limitations of data processing and analysis

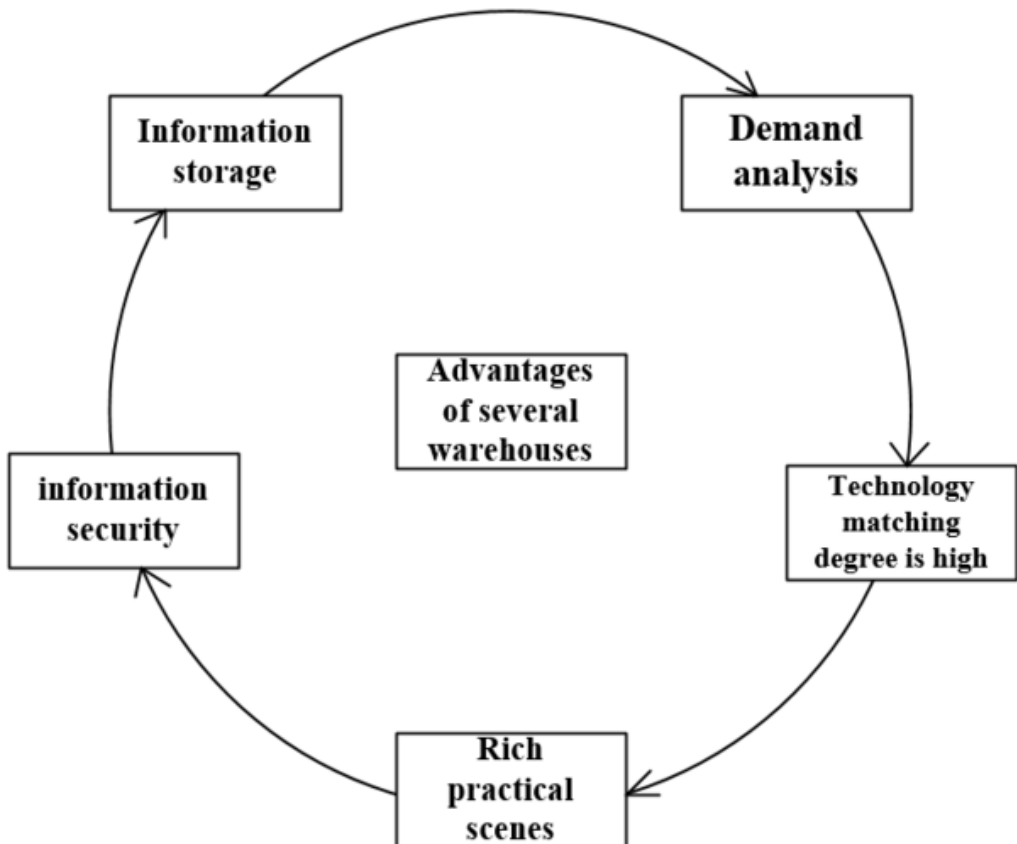
ability, the traditional educational evaluation system often cannot provide timely, accurate, and comprehensive decision support for educators, learners, and managers. As a result, educational decision-making is often based on experience or intuition, lacking scientificity and objectivity (Webber & Zheng, 2020).

In view of the above problems, there is an urgent need to find a more efficient, intelligent, and comprehensive solution to optimize education evaluation systems. As a subject-oriented, integrated, relatively stable, and time-varying method of data collection, a data warehouse can not only solve the problems of scattered data and low processing efficiency but also provide more accurate and comprehensive data support for educational evaluation through its powerful data processing and analysis capabilities. Therefore, it is a feasible solution to apply a data warehouse to the education evaluation system.

Potential Analysis of Several Warehouses as a Solution

In the face of the current challenges faced by educational evaluation systems, a data warehouse, as an advanced data management and analysis technology, shows great potential as a solution (Dibouliya et al., 2023). In this paper, the application potential of data warehouses in the education evaluation system were analyzed by combining the advantages of data warehouses shown in Figure 2.

Figure 2. Advantages of several warehouses



First, a data warehouse can solve the problem of data dispersion and islands. In the field of education, data are often scattered in different departments, systems, and platforms, making it difficult to form effective data sharing and circulation. A data warehouse, through its powerful data integration ability, can store and manage data from different channels and formats in a unified way and realize data centralization and standardization. This not only solves the problem of data dispersion but also breaks the information island, enabling data to be shared and circulated within educational institutions and improving the efficiency and value of data utilization (Li et al., 2022).

Second, the data warehouse has efficient data processing and analysis capabilities. Compared with traditional data processing methods, a data warehouse adopts more advanced data processing technologies and algorithms, which can realize rapid processing and analysis of massive data (Jameel et al., 2022). In the educational evaluation system, a data warehouse can use these technologies to explore and analyze the multidimensional data of students' academic performance, comprehensive quality, and behavior performance, and find the correlation and regularity between the data, so as to provide more accurate and comprehensive data support for educational evaluation.

Thirdly, a data warehouse can provide powerful decision support functions. In the field of education, decision-making often depends on in-depth analysis and understanding of data. The data warehouse, through its powerful data analysis and visualization capabilities, transforms complex data into intuitive charts and reports, helping educators and learners better understand the meaning and trends behind the data, so as to make more scientific and reasonable decisions. While improving the quality and efficiency of educational decision-making, it can also promote the in-depth development of educational reform (Rao, 2024).

Fourth, the data warehouse has strong scalability and flexibility. With the continuous growth and change of educational data, a data warehouse can expand its storage and processing capacity as needed to meet the new data requirements (Maswanganyi et al., 2024). At the same time, it also supports a variety of data models and query methods, which can meet the changes of different users and data needs. This scalability and flexibility enable the data warehouse to continuously adapt to the development needs of the educational evaluation system and provide strong support for the digital transformation in the education field.

To sum up, the data warehouse, as an effective solution to the challenges facing the educational evaluation system, has great potential and value. By giving full play to its advantages in data integration, processing, analysis and decision support, more efficient, intelligent and comprehensive solutions can be provided for data management and analysis in the field of education, thus promoting the continuous optimization and upgrading of the educational evaluation system.

RESULTS

The application of a data warehouse in the educational evaluation system is not only an innovation of the traditional evaluation mode but also an important force to promote education modernization and intelligence. Next, this paper will discuss the specific application of data warehouse in the educational evaluation system in detail.

Data Integration and Management

The primary application of a data warehouse is to integrate educational data from different sources and formats. By constructing a unified data model and data warehouse architecture, multi-source data are stored and managed centrally. Let us take the data shown in Figure 3 and Figure 4 as an example to illustrate.

Figure 3. Data decentralized storage state

Table 1-1: Students' Academic Achievements		
Student ID	Subject	Score
1	Math	90
2	Chinese	85
1	English	88

Table 1-4: Teacher evaluation		
Student ID	Teacher ID	Evaluation content
1	T001	Learning attitude is serious and progress is remarkable
2	T002	Need to strengthen classroom participation

Table1-2: Student behavior record		
Student ID	Date	Behavior description
1	XXXX/X/X	Actively participate in class discussions
2	XXXX/X/X	Late

Table 1-5: Parental feedback		
Student ID	Parent ID	Feedback content
1	P001	Children are active at home and willing to learn.
2	P002	It is hoped that schools will pay more attention to children's mental health.

Table 1-3: Evaluation of students' comprehensive quality			
Student ID	Morality	Teamwork	Innovation capacity
1	A	B	A
2	B	A	B

Data decentralized storage state

Figure 4. State after centralized storage and management of data

Student ID	Name	Subject	Score	Morality	Teamwork	Innovation capacity	Date	Behavior description	Teacher ID	Evaluation content	Parent ID	Feedback content
1	San Zhang	Math	90	A	B	A	XXXX/X/X	Actively participate in class discussions	T001	Learning attitude is serious and progress is remarkable	P001	Children are active at home and willing to learn
2	Si Li	Chinese	85	B	A	B	XXXX/X/X	Late	T002	Need to strengthen classroom participation	P002	It is hoped that schools will pay more attention to children's mental health
1	San Zhang	English	88	A	B	A	NULL	NULL	NULL	NULL	NULL	NULL

State after centralized storage and management of data

In the state of scattered data storage, all kinds of information about students are separated and stored in different tables. This decentralized data storage mode not only makes data query and integration complicated but also increases the difficulty of data updating and maintenance, which easily leads to data inconsistency or omission. In addition, this storage mode also greatly limits the potential of data analysis and

mining, making it difficult to obtain comprehensive and accurate data support for educational evaluation.

In contrast, when data is stored and managed centrally in a unified data model and data warehouse architecture, these challenges are solved. Under this framework, all the information about students is integrated into a table, and the complete information of students can be easily obtained through keywords such as student ID. This not only greatly simplifies the process of data query and integration but also improves the efficiency and accuracy of data update and maintenance. More importantly, the centrally stored data provides rich materials for data analysis and mining, which enables educational evaluation to make more scientific and reasonable judgments based on more comprehensive and accurate data.

Multidimensional Evaluation

The data warehouse supports multidimensional and deep-seated mining and analysis of educational data. By constructing an evaluation model, students' academic performance, comprehensive quality, hobbies, social skills, and other dimensions can be comprehensively evaluated.

Figure 5. Multidimensional data analysis using python

```
import pandas as pd
# Dataset
data={
    Student ID:[1,2,3,4],
    Academic achievements: [85,90,78,88],
    Comprehensive quality: [80,85,70,95],
    Hobbies: [90,85,88,92],
    Social skills: [75,80,70,85]
}
# Create DataFrame
df = pd.DataFrame(data)

# Calculate the comprehensive evaluation score
df[ 'comprehensive evalustion']=(
df[ 'Academic achievements' ]*weights[ 'Academic achievements' ]+
df[ 'Comprehensive quality' ]*weights[ 'Comprehensive quality' ]+
df[ 'Hobbies' ]*weights[ 'Hobbies' ]+
df[ 'School skills' ]*weights[ 'School skills' ]+
)
# Output results
```

As Figure 5 shows, pandas in Python is used for multidimensional data analysis. This evaluation method is not only more comprehensive and objective but also reveals the differences and uniqueness among students. Schools can also use data warehouse technology to statistically analyze students'

academic achievements, find out their strengths and weaknesses, and then provide students with personalized learning suggestions and counseling. At the same time, through the comprehensive quality evaluation model, the school can more accurately evaluate the all-round development of students, provide a scientific basis for students' growth planning, and provide strong support for personalized teaching.

Personalized Learning Support

The powerful data analysis ability of a data warehouse has brought unprecedented personalized learning support to the field of education. By digging deep into multidimensional data such as students' learning trajectory, learning style, and learning effectiveness, each student's learning needs and preferences can be understood more accurately.

Specifically, the data warehouse can record and analyze students' learning progress in different courses and different time periods, as well as their reactions and solutions when they encounter problems. Based on these data, the learning materials and exercises can be tailored to individual students. For students with faster learning progress, more challenging learning content can be recommended to stimulate their exploration spirit; for students who encounter bottlenecks, targeted counseling materials and exercises can be provided to help them overcome difficulties and improve.

In addition, the data warehouse can also capture students' hobbies and learning preferences. By analyzing students' learning performance in different fields, customized learning paths and course recommendations can be provided. For example, students who like scientific experiments can be recommended more courses and practical activities related to experiments; students who are keen on literary creation can be provided with rich literary resources and creative guidance.

This kind of personalized learning support can not only stimulate students' interest in learning, but also make them participate in learning more actively, thus significantly improving learning efficiency. Students can master knowledge more quickly and improve their academic performance after getting suitable learning resources and suggestions.

Optimization of Educational Decision-Making

The data warehouse provides rich data support for educational administrators, helping them formulate educational policies and plans more scientifically and accurately. As shown in Table 1, in-depth analysis of educational data can provide insight into the distribution of educational resources, students' learning effectiveness, teachers' teaching quality and other key information, thus providing strong data support for educational decision-making.

Table 1. Overview of education data analysis

School/region	Number of teachers	Number of students	Student-teacher ratio	Teaching facilities score	Teaching quality score	Average grade of students	Excellent rate	Pass rate
School A	65	1300	20	92	88	85	25%	95%
School B	50	1050	21	85	82	80	20%	90%
School C	70	1500	21.43	95	92	90	30%	98%
School D	45	900	20	78	78	75	15%	85%
Area E	-	4500	-	-	-	80	22%	92%
Area F	-	3800	-	-	-	78	18%	88%

continued on following page

Table 1. Continued

School/region	Number of teachers	Number of students	Student–teacher ratio	Teaching facilities score	Teaching quality score	Average grade of students	Excellent rate	Pass rate
School G (special education)	20	300	15	90 (customized facilities)	90 (professional assessment)	70 (special standards)	-	90% (special standard)
School H (international school)	80	800	10	98 (international facilities)	95 (international assessment)	92 (international courses)	35%	99%

The differences and complexity of different schools or regions in the allocation of educational resources, students’ learning effectiveness, and teaching quality can be fully understood. For example, for school G (special education), the data reflect the characteristics of special education schools, including customized facilities and special evaluation standards. As far as the overall evaluation of teaching quality is concerned, the quantitative evaluation of teaching quality is made by integrating many factors, such as students’ grades, students’ satisfaction with teachers, innovation of teaching methods, and so forth. If we want to know something about this school district, then Region E and Region F can be taken as examples. The data are summary data that fully show the average situation of all schools in these regions. These data can provide more abundant and in-depth data support for educational decision-making and help decision-makers formulate educational policies and plans more scientifically.

DISCUSSIONS

Although the application of data warehouse in the education evaluation system has brought many conveniences and advantages, there are also some potential disadvantages. The following are the main disadvantages that may exist.

Data Quality Issues

The data in the data warehouse comes from many different educational information systems, such as student information management systems and teacher management information systems. Because there may be differences in data standards and formats between different systems, there may be inconsistencies, inaccuracies, or omissions in the data integration process. These problems will lead to the decline of data quality, thus affecting the accuracy and reliability of educational evaluation.

Data Update Delay

The data in the data warehouse usually needs to be processed and transformed before they can be used for educational evaluation. There may be a certain time delay in this process, resulting in the data not reflecting the current education situation in real time. This may make decision-makers unable to obtain the latest educational information in time, thus leading to inaccurate judgments.

Data Security Issues

A large amount of educational data is stored in the data warehouse, including sensitive information such as students’ personal information and academic performance. If security measures are not in place, it may lead to data leakage or malicious use. This will not only harm students’ privacy rights but also have a negative impact on the fairness and authority of educational evaluation.

Excessive Dependence on Data

Although data play an important role in educational evaluation, over reliance on data may ignore other important evaluation factors. For example, factors that are difficult to quantify, such as students' comprehensive quality and teachers' teaching style, may not be fully reflected by data. Therefore, many factors should be considered comprehensively in the educational evaluation to avoid the one-sidedness of the evaluation results caused by excessive dependence on data.

To sum up, the application of a data warehouse in the educational evaluation system needs to be done with caution about its potential disadvantages and with effective measures to prevent and solve them.

CONCLUSIONS

The application of data warehouse in the education evaluation system significantly improves the accuracy and comprehensiveness of evaluation and provides a more scientific and objective decision-making basis for educators, learners, and managers. However, the potential disadvantages, such as data quality, data update delay, data security, and over reliance on data, cannot be ignored. In the future, with the continuous progress of technology and in-depth application, further measures should be taken to improve the construction and management of the data warehouse system, improve data quality and processing efficiency, and strengthen data security measures. At the same time, there should be close attention to the comprehensive evaluation factors to avoid the one-sidedness of the evaluation results caused by excessive dependence on data. In the future, data warehouses will play a more important role in promoting the modernization and intelligence of education.

COMPETING INTERESTS

The authors of this publication declare there are no competing interests.

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